328652(28)

B. E. (Sixth Semester) Examination, Nov.-Dec. 2019

(New Scheme)

(ET & T Engg. Branch)

ELECTRONIC CIRCUIT DESIGN

Time Allowed : Three hours

Maximum Marks : 80

Minimum Pass Marks : 28

Note *: Attempt all the questions. Part (a) of each question is compulsory and carries 2 marks.. Attempt any* ***two*** *from (b), (c) & (d) and carries 7 marks.*

Unit-I

1. (a) Define Sweep time.

(b) The diodes are ideal. Plot V_0 against V_i

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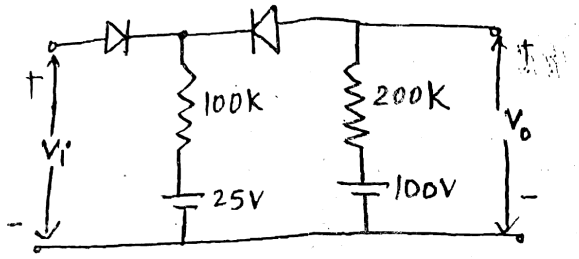
(Given the voltage V_i varies from 0 to 150 V)



(c) Explain square wave generator circuit using Opamp. *Prove that its frquency* of oscillation is given by :

$$f_0 = \frac{1}{2R_C}$$

(d) State and prove the clamping circuit theorem.

Unit-II

2. (a) Define Bistable multivibrator.

(b) Explain collector coupled monostable multivibrator and derive its expression for gate width. Also draw all the relative waveforms.

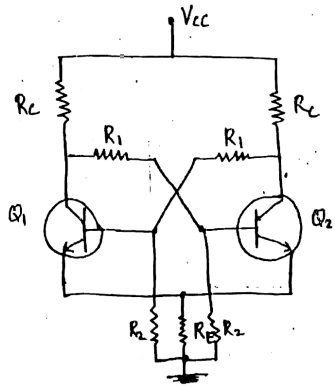
(c) The self biased binary using n-p-n silicon transistors having worst case (maximum) values of $V_{CE}(sat)$ = 0.4 V, $V_{BE}(sat)$ = 0.8 V, and zero base to emitter voltage for cut off. The circuit parameters are V_{CC} = 20 V.

$R_C = 4{\cdot}7\ K$, $R_1 = 30\ K$, $R_2 = 15\ K$ and $R_E = 390\,\Omega$

(i) Find the stable state currents and voltages.

(ii) Find the minimum value of h_{FE} required to give the values in part (a).

(iii) As the temperature is increaed, what is the maximum value to which I_{CBO} can increase before the condition is reached where neither transistor is OFF.



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(d) Design an astable multivibrator using transistor to generate a square wave of 1 kHz frequency and duty cycle of 25%. Take $V_{CE}(sat) = 0.3$ V, $V_{BE}(sat) = 0.7$ V and $I_{B(actual)} = 1.5\ I_{B(min)}$, when the transistor is in saturation $I_{C(sat)} = 10$ mA are $V_{CC} = +12$ *V*. (Take $(h_{Fe})_{min} = 40$)

Unit-III

3. (a) Why 555 timer is called so?

(b) *Explain astable* multivibrator using IC555. Also derive the relation for its duty cycle.

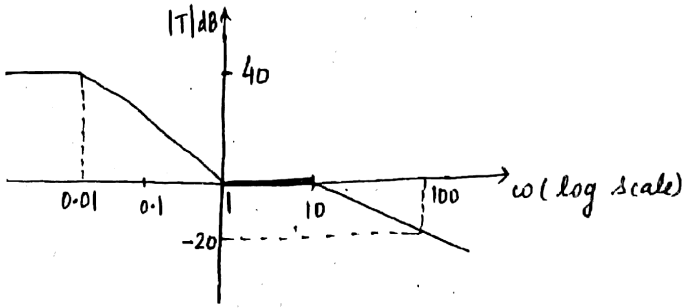
(c) Design a missing pulse detector using IC 555.

(d) Design a 555 astable multivibrator with an output signal frequency of 700 Hz and 50% duty cycle.

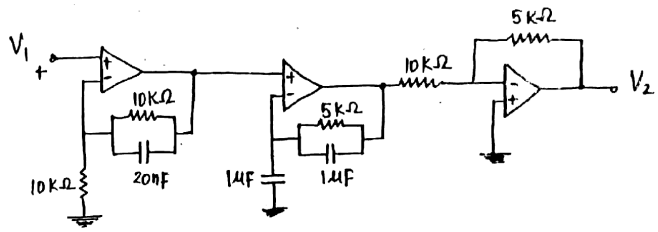
Unit-IV

4. (a) Define Bilinear Transfer function.

(b) The transfer function corresponding to the asymptotic Bodeplot shown has only real poles and zeros construct T(S), determining all constants.



(c) For the circuit shown, prepare an asymptotic bode plot for the magnitude of T(jw) :



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(d) Explain low pass filter using 3 OP-amp biquad circuit. Derive its transfer function.

Unit-V

5. (a) Give the standard form of transfer function for 2nd order low pass circuit.

(b) Design Butterworth filter using Sallen and key circuit design 3 :

Take :

$\alpha_{max} = 0 \cdot 5\,dB$

$\alpha_{min} = 30\,dB$

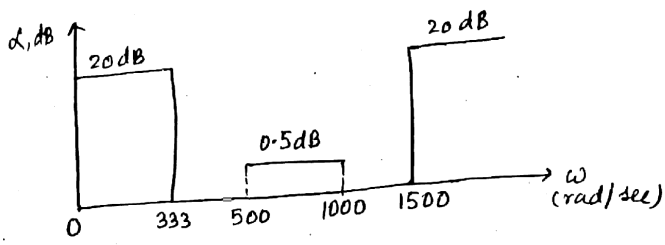
$k = 1$

$W_p = 750$

$W_s = 1750$

use scaling in final design.

(c) Design a Chebyshev band pass filter from the attenuation specification diagram :



(d) Explain sensitivity of Sallen and key circuit.

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